Research Meeting

2026-08-24

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Research to date

Background	**EHRs contain extensive information, while data needed for care is spread across multiple screens**
Goal	**Bring together the information required for each clinical task and support efficient review**
Approach	**Identify the required data and presentation from the task, then create a UI suited to its purpose**

Change how clinicians reach the information they need

Fixed screens

Required information is spread across screens and records.

Users must remember where it is and how to reach it.

Task-aligned screens

The system gathers the data needed for the question at hand.

Lists, tables, and trends appear together for the same clinical context.

Recent progress and CHI 2027 submission

User evaluation

1 Discussing potential collaborators through Dr. Yabe

2 Starting after ethics and study requirements are met

CHI 2027 Papers

1 Build evaluation data from EHRSQL, which pairs clinical questions with reference SQL

2 Detect, locate, and repair gaps across questions, SQL, query results, and screens

The first submission will focus on technical evaluation; human-subject evaluation will follow as future validation

Connect tasks, data, and presentation to build the UI

Clinical question

Organize the user's task and clinical context.

Task graph

Connect clinical tasks, required data, and interface components.

UI with data

Run SQL and render the results as tables or charts.

Current implementation and future generation

Current

We manually create structural JSON from interviews.

The system reads the task graph and renders the UI with rules.

Future

A custom model will interpret the user's request.

It will construct the proposed task graph and update the UI.

Preoperative anesthesia scenario developed with Dr. Ito

We organized the clinical setting and review tasks based on interviews with Dr. Ito

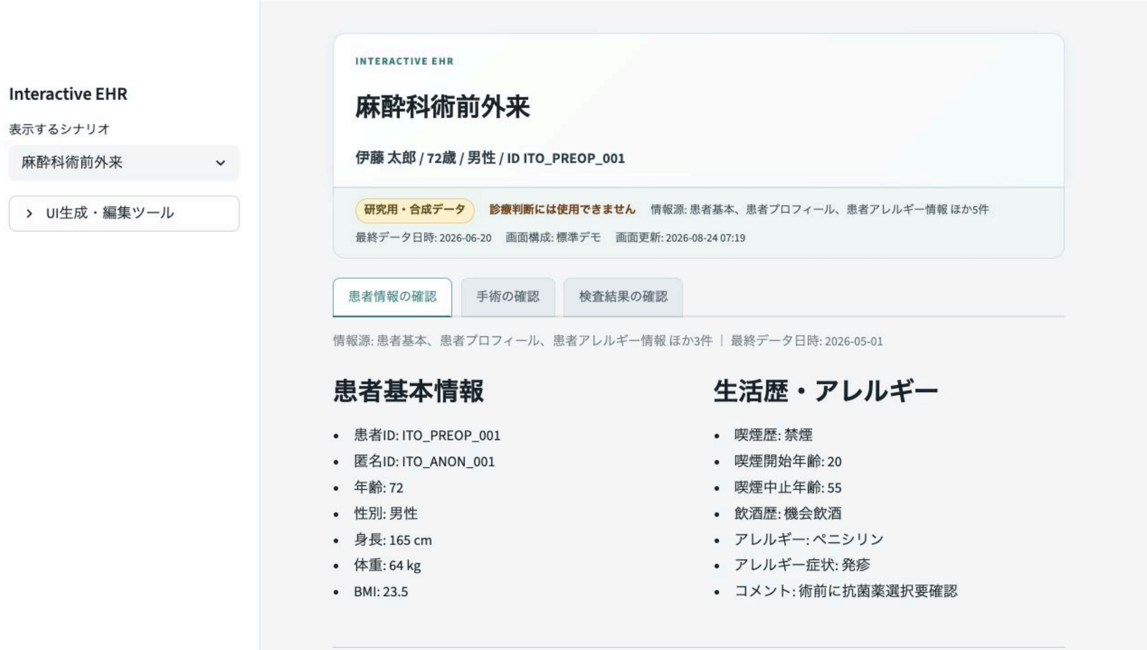
Clinical setting

Before surgery, review the patient's background and prepare anesthesia precautions and explanations

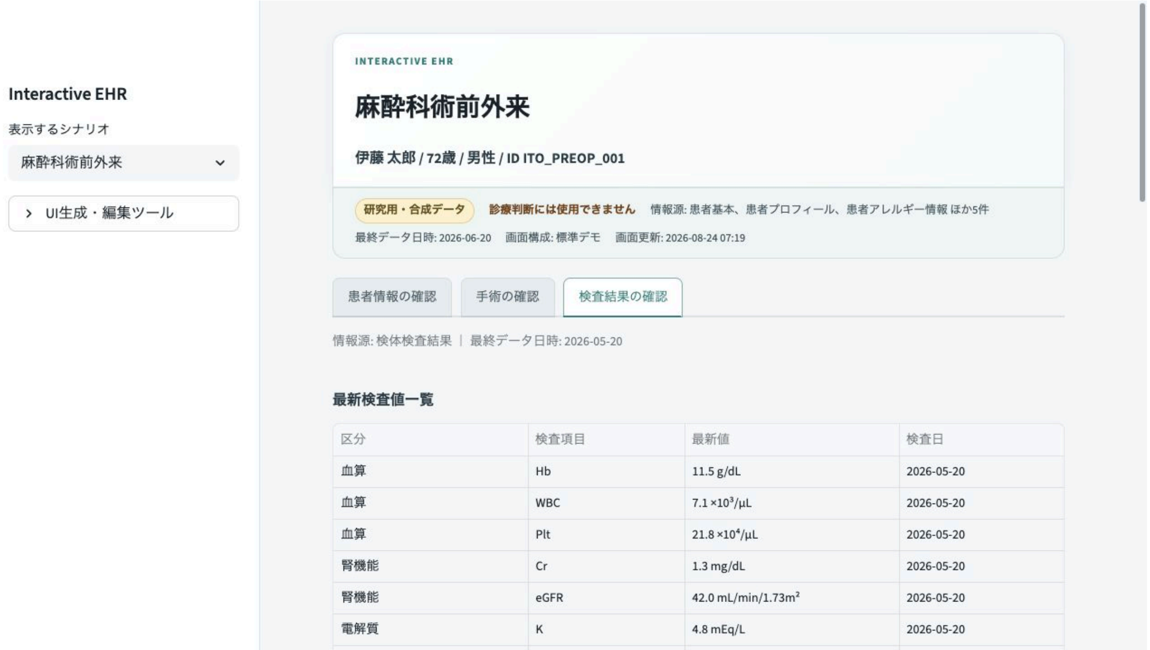
Review tasks

- Patient background and planned surgery
- Social history and allergies
- Medical history, treatment, and laboratory results
- Anesthesia history and anesthesia risks

Preoperative anesthesia clinic UI



Patient information



Laboratory results

Synthetic data only. The patient and timestamp remain fixed while the review target changes

Compare screens using the same DWH data

Hospital DWH

Use the same data referenced by the operational EHR.

Traceable retrieval

Trace displayed values to SQL, source tables, and data timestamps.

UI comparison

Compare the current screen and proposed UI using the same data.

Planned user evaluation

Comparison design

1. Matched synthetic cases
2. Within-participant EHR vs. UI
3. Counterbalanced order

Measures

1. Success and critical omissions
2. Time, actions, and screen transitions
3. NASA-TLX, SUS, interviews

Begin after expert review, ethics approval, and institutional authorization

Feedback requested from clinicians

Review it as a clinical interface

- Does it include the information needed to prevent critical omissions?
- Are the information groups and review order natural?
- Can the source, timestamp, and retrieval conditions support verification?
- Are the comparison conditions with the current EHR fair?